SalesSpectrum – Retail Sales Analysis using SQL

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Introduction

SalesSpectrum – Retail Sales Analysis using SQL

- SalesSpectrum is a SQL-based project focused on analyzing retail sales data.
- The dataset includes product details, monthly demand, pricing, and total sales value
- The project uses SQL queries to extract meaningful insights from structured data
- It helps in understanding product performance and revenue trends

Objective

- To analyze product-level sales performance
- To identify top and low-performing products
- To **evaluate category-wise** revenue distribution
- To analyze monthly demand trends
- To apply SQL concepts like aggregation, sorting, subqueries, and grouping

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Problem Statements

Businesses face difficulty in:

- Identifying high-revenue products
- Tracking monthly demand patterns
- Understanding category performance
- Making data-driven decisions

Raw data does not provide insights directly

Solutions

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1. Product Performance Analysis
Identified **Top 5 revenue-generating products** → Key revenue drivers
Identified **Bottom 5 low-performing products** → Require improvement or removal

2. Pricing vs Sales Analysis
Found **highest priced product** → Evaluated impact of price on sales performance
Calculated **average price per category** → Compared pricing across categories

3. Category-Level Insights
Calculated **category-wise total revenue** → Identified top contributing categories
Compared **product performance within categories** → Determined most profitable segments

4. Advanced Analysis
Identified **products with above-average sales** → Strong performers
Analyzed **monthly demand (Jan–Dec)** → Observed trends and seasonality

Outputs

1. Top 5 Revenue Generating Products

Item_Name	Total_Sales_Value
Ten And	59462000
Radio Race	47606000
Pattern Book	37136000
Animal Key	29804500
Material Vote	28431000

2. Bottom 5 Products (Low Performance)

Item_Name	Total_Sales_Value
Gun Both	1228
Worry Because	1290
Indicate Organization	1396
Along Have	1456
Investment Agreement	1512

3. Category-wise Revenue

Category	Revenue
Grocery	911737236
Apparel	99381360
Electronics	25081150
Toys	21945148
Home & Kitchen	14143006

4. Highest Priced Product

Item_Name	Price_Per_Unit
Few Think	1000

5. Average Price per Category

Category	Avg_Price
Grocery	65.7811
Apparel	57.8241
Toys	63.2360
Electronics	44.8705
Home & Kitchen	41.3600

6. Create View for Category Summary

Category	Revenue
Grocery	911737236
Apparel	99381360
Toys	21945148
Electronics	25081150
Home & Kitchen	14143006

Insights

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1. Identified Top 5 revenue-generating products
2. Detected Bottom 5 low-performing products
3. Found category-wise revenue contribution
4. Calculated average price per category

5. Identified highest priced product
6. Found products with above-average sales performance
7. Observed monthly demand patterns across the year

These insights help businesses improve strategy and decision-making

Future Scope

I. Integrate SQL results with Power BI for visualization
II. Add real-time sales data for live analysis
III. Perform predictive analysis using machine learning
IV. Expand dataset with customer and regional data
V. Build dashboards for business reporting

Conclusion

- Successfully built a SQL-based retail analysis system
- Converted raw data into actionable insights
- Demonstrated strong SQL skills (aggregation, grouping, subqueries)
- Enabled better understanding of product and revenue performance

SalesSpectrum provides a complete overview of sales and demand trends

Thank You